Biological Data Analysis Term

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1 Problem 1

We only need to condition on L_1, as doing so ensures that L_1, L_2, and L_3 are no longer confounders between A and Y. In other words, conditioning on L_1 closes all open paths that could generate spurious associations between A and Y. As a result, the remaining observed association reflects the causal effect of A on Y.

2 Problem 2

1. Explain the identifying assumptions required to estimate the average causal effect in plain words.

There are three identifying assumptions required to estimate the average causal effect in plain words. For simplicity, we denote Y as `sal07`, A as `prog`, and L as the vector representing the remaining variables in the dataset.

1. Consistency assumption: $Y(A) = A$. This assumption implies that for each citizen, their annual salary for the following year, if they had participated in the program, would be the same as their actual salary if they did participate. Conversely, their annual salary for the following year, if they had not participated in the program, would be the same as their actual salary if they did not participate.
2. Conditional Exchangeability assumption: $Y(A) \perp A \,|\, L$. This assumption implies that there are no confounders between A and Y after conditioning on L. To elaborate, after conditioning on L, the unobserved potential outcome $Y(0)$ for those who participated in the program has the same distribution as $Y(0)$ for those who did not participate, allowing us to identify the causal effect for those who participated in the program. Similarly, the same logic applies to identifying the causal effect for those who did not participate in the program. The intuition provided thus explains why the average causal effect can be identified with this key assumption.
3. Positivity assumption: $p(A = a|L) > 0$ for $a = 0, 1$. This assumption implies that all possible values of A can be observed under each stratification of L.

2. Fit a propensity model that describes the citizens' choice to enroll in the program. Argue if the causal query made by the city administrator is answerable by the data or not. If not, advise how the city administrator may reframe his query.

Suppose the consistency and the conditional exchangeability assumptions are satisfied, it remains to check if the positivity assummption is satisfied so that the average causal effect can be identified. To check the positivity assumption we fit the propensity model with a logistic regression as follows.

$$\text{logit}P(A = 1|L) = \beta_0 + \beta^T b(L),$$

where $b(L)$ is a vector of basis functions, including the dummy variables for `female`, `race`, and the restricted cubic splines of $k = 4$ for `age`, `educ` and `sal05`.

```r
dat = read.csv("salary.csv")
dat$female = as.factor(dat$female)
dat$race = as.factor(dat$race)
fit1 = glm(prog~rcs(age, 4)+rcs(educ, 4)+rcs(sal05, 4)+female+race, family = "binomial", data = dat)
summary(fit1)
```

```
##
## Call:
## glm(formula = prog ~ rcs(age, 4) + rcs(educ, 4) + rcs(sal05,
##     4) + female + race, family = "binomial", data = dat)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)          -1.511e+01  7.590e-01 -19.912  < 2e-16 ***
## rcs(age, 4)age        5.704e-01  3.290e-02  17.334  < 2e-16 ***
## rcs(age, 4)age'      -2.335e+00  1.709e-01 -13.667  < 2e-16 ***
## rcs(age, 4)age''      4.557e+00  3.513e-01  12.973  < 2e-16 ***
## rcs(educ, 4)educ      9.494e-02  1.886e-02   5.035 4.78e-07 ***
## rcs(educ, 4)educ'     6.318e-03  2.080e-02   0.304 0.761280
## rcs(sal05, 4)sal05   -5.389e-04  4.411e-05 -12.215  < 2e-16 ***
## rcs(sal05, 4)sal05'   1.156e-02  1.477e-03   7.830 4.87e-15 ***
## rcs(sal05, 4)sal05'' -1.235e-02  1.600e-03  -7.719 1.17e-14 ***
## female1              -5.744e-02  5.605e-02  -1.025 0.305434
## raceHispanic          7.886e-01  1.388e-01   5.683 1.33e-08 ***
## raceOthers            3.815e-01  1.078e-01   3.537 0.000404 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 10203.8  on 10271  degrees of freedom
## Residual deviance:  8060.1  on 10260  degrees of freedom
## AIC: 8084.1
##
## Number of Fisher Scoring iterations: 6
```

As seen in the summary table, we remove `female` and the nonlinear terms of `educ` from the model as they appear not significant according to the Wald's test. The refitted model is shown below.

```r
fit1 = glm(prog~rcs(age, 4)+educ+rcs(sal05, 4)+race, family = "binomial", data = dat)
summary(fit1)
```

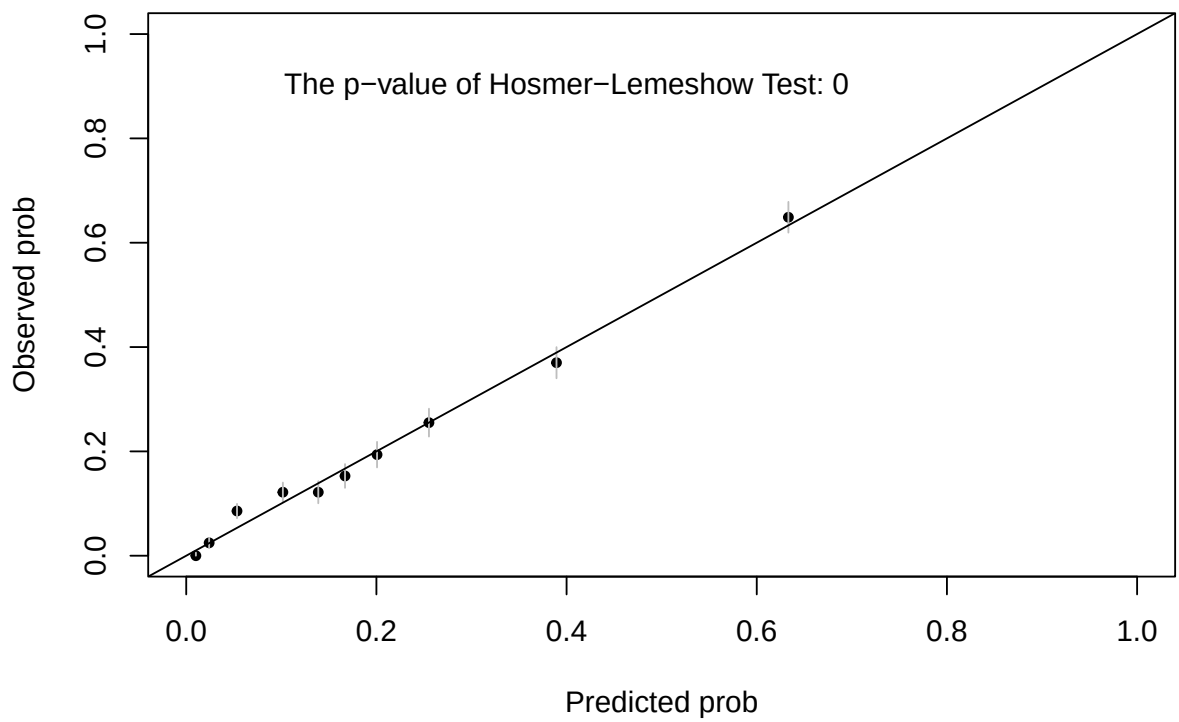
```
##
## Call:
## glm(formula = prog ~ rcs(age, 4) + educ + rcs(sal05, 4) + race,
##     family = "binomial", data = dat)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)          -1.522e+01  7.200e-01 -21.138  < 2e-16 ***
## rcs(age, 4)age        5.719e-01  3.263e-02  17.524  < 2e-16 ***
## rcs(age, 4)age'      -2.342e+00  1.698e-01 -13.795  < 2e-16 ***
```

```
## rcs(age, 4)age''      4.571e+00  3.493e-01  13.089  < 2e-16 ***
## educ                  9.990e-02  9.851e-03  10.141  < 2e-16 ***
## rcs(sal05, 4)sal05   -5.378e-04  4.408e-05 -12.199  < 2e-16 ***
## rcs(sal05, 4)sal05'   1.153e-02  1.475e-03   7.813 5.57e-15 ***
## rcs(sal05, 4)sal05'' -1.231e-02  1.599e-03  -7.702 1.34e-14 ***
## raceHispanic          7.945e-01  1.378e-01   5.764 8.24e-09 ***
## raceOthers            3.807e-01  1.078e-01   3.530 0.000415 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 10203.8  on 10271  degrees of freedom
## Residual deviance:  8061.2  on 10262  degrees of freedom
## AIC: 8081.2
##
## Number of Fisher Scoring iterations: 6
```

To check the adequacy of the current model, we make a calibration plot of the model and examine its calibration using Hosmer-Lemeshow test.

```r
plot.calibration = function(dat, fit){
  propensities = predict(fit, type = "response")
  dat$fitted = propensities
  dat$group = cut(dat$fitted,
  quantile(dat$fitted, (0:10)/10),
  include.lowest = T)
  hl.dat = dat %>%
  group_by(group) %>%
  dplyr::summarize(sum_prob = sum(fitted), sum_non_prob = sum(1-fitted),
  sum_event = sum(prog == 1), sum_non_event = sum(prog == 0),
  mean_prob = mean(fitted), mean_event = mean(prog == 1),
  n = n()) %>%
  mutate(se = sqrt(mean_prob*(1-mean_prob)/n),
  lower = pmax(0, mean_event-1.96*se),
  upper = pmin(1, mean_event+1.96*se))
  plot(hl.dat$mean_prob, hl.dat$mean_event, xlim = c(0, 1), ylim = c(0, 1),
  xlab = "Predicted prob", ylab = "Observed prob", pch = 20)
  for(i in 1:nrow(hl.dat)){
  points(hl.dat$mean_prob[c(i, i)], c(hl.dat$lower[i], hl.dat$upper[i]),
  col = "grey", type = "l")
  }
  abline(a = 0, b = 1)

  hl = sum((hl.dat$sum_event - hl.dat$sum_prob)^2/hl.dat$sum_prob) +
  sum((hl.dat$sum_non_event - hl.dat$sum_non_prob)^2/hl.dat$sum_non_prob)
  p.value = 1-pchisq(hl, 8)

  text(0.4, 0.9, paste("The p-value of Hosmer-Lemeshow Test:", round(p.value, 4)))
}
```

```r
plot.calibration(dat, fit1)
```



We find that this model does not seem adequate as the p-value of the Hosmer-Lemeshow test is smaller than 0.05. As a result, we try to increase the extent of non-linearity of `age` and `sal05` in the model. Specifically, we set the restricted cubic splines of $k = 6$ for `age`, and set the restricted cubic splines of $k = 5$ for `sal05`. The refitted model is shown below.

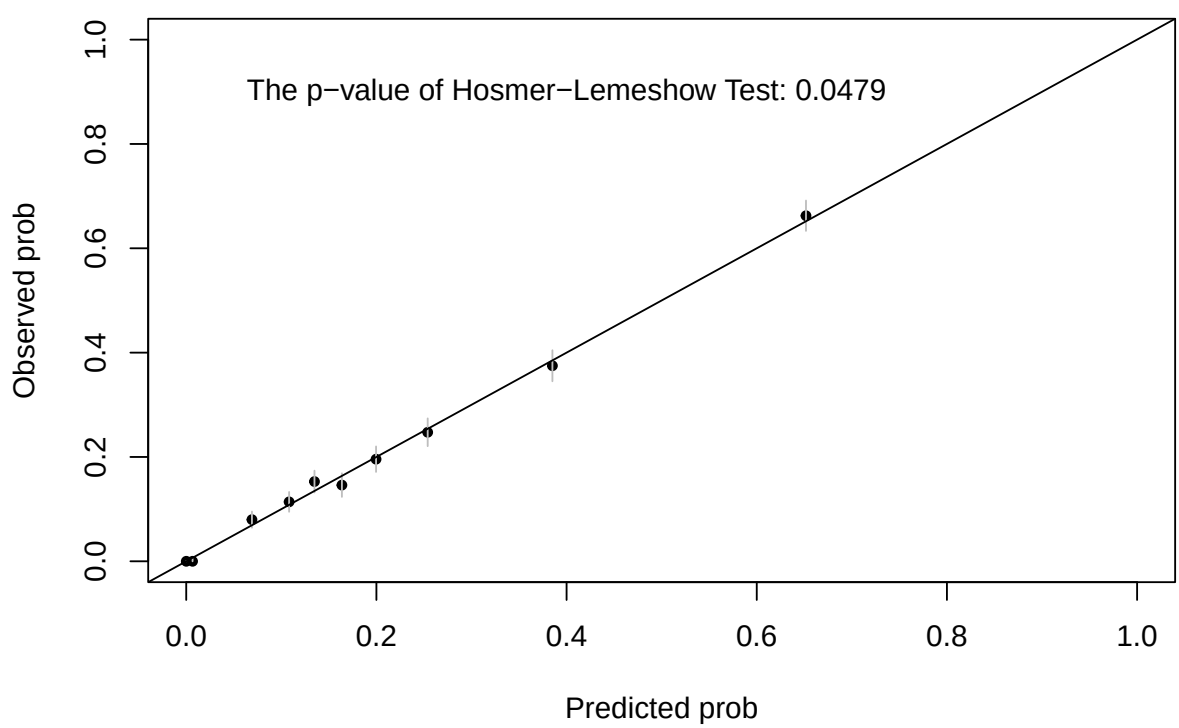
```
fit1 = glm(prog~rcs(age, 6)+educ+rcs(sal05, 5)+race, family = "binomial", data = dat)
summary(fit1)
```

```
##
## Call:
## glm(formula = prog ~ rcs(age, 6) + educ + rcs(sal05, 5) + race,
##     family = "binomial", data = dat)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)           -5.730e+01  5.527e+00 -10.368  < 2e-16 ***
## rcs(age, 6)age         2.696e+00  2.720e-01   9.911  < 2e-16 ***
## rcs(age, 6)age'       -3.917e+01  4.268e+00  -9.178  < 2e-16 ***
## rcs(age, 6)age''       8.483e+01  9.552e+00   8.881  < 2e-16 ***
## rcs(age, 6)age'''     -5.208e+01  6.429e+00  -8.101 5.47e-16 ***
## rcs(age, 6)age''''     7.257e+00  1.542e+00   4.706 2.53e-06 ***
## educ                   9.900e-02  9.922e-03   9.978  < 2e-16 ***
## rcs(sal05, 5)sal05    -9.908e-04  7.496e-05 -13.218  < 2e-16 ***
## rcs(sal05, 5)sal05'    3.849e-02  3.932e-03   9.788  < 2e-16 ***
## rcs(sal05, 5)sal05''  -4.313e-02  4.458e-03  -9.673  < 2e-16 ***
```

```
## rcs(sal05, 5)sal05'''  5.944e-03  7.492e-04   7.933 2.14e-15 ***
## raceHispanic           8.166e-01  1.390e-01   5.876 4.20e-09 ***
## raceOthers             3.961e-01  1.086e-01   3.649 0.000263 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 10204  on 10271  degrees of freedom
## Residual deviance:  7852  on 10259  degrees of freedom
## AIC: 7878
##
## Number of Fisher Scoring iterations: 9
```

To check the adequacy of the current model, we make a calibration plot and examine its calibration using Hosmer-Lemeshow test.

```
plot.calibration(dat, fit1)
```

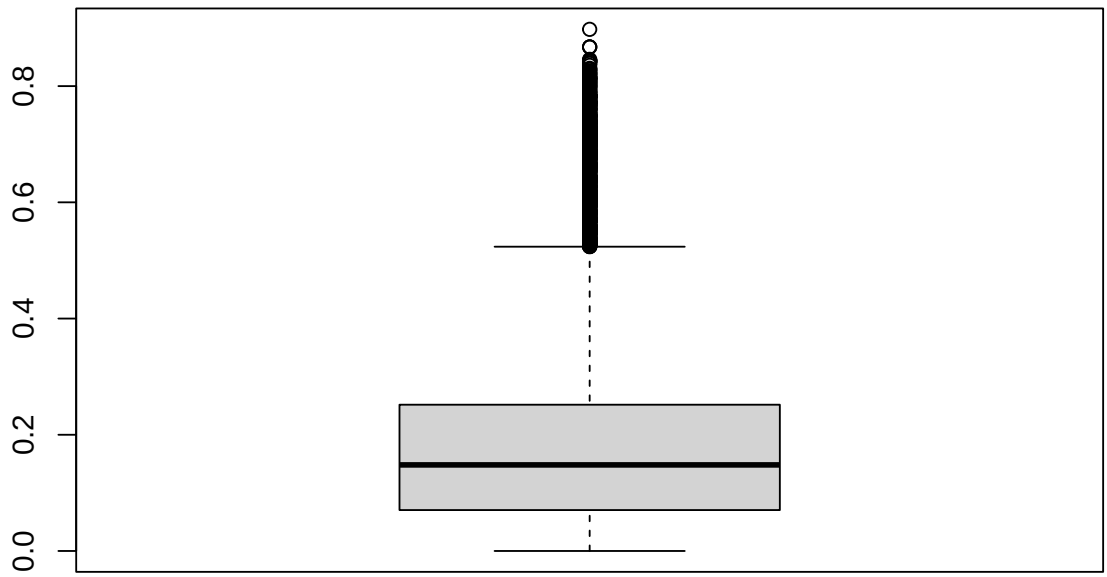



```
propensities = predict(fit1, type = "response")
```

Though the p-value is 0.0479, slightly smaller than 0.05, the model should be considered satisfactory at this stage. (Honestly, despite extensive trials and errors, I was unable to identify a model that produced a p-value greater than 0.05 in the Hosmer-Lemeshow test.) We then make the boxplot of the predicted propensities

based on the current model. We find that the minimum and maximum values of the predicted propensities are 3.0402566×10^{-7} and 0.8977198 respectively, where the former may be too small. Therefore, we conclude that the positivity assumption may not hold for this data. With the lack of the positivity assumption, the causal query made by the city administrator is not answerable.

```
boxplot(propensities)
```




```
c(min(propensities), max(propensities))
```

```
## [1] 3.040257e-07 8.977198e-01
```

I would advise the city administrator to reframe their query to focus on the causal effect for individuals with at least a 3% probability of participating in the program. By addressing this refined query, the administrator can still use the analysis results to persuade those who are very reluctant to participate in the program to consider joining, though sacrificing approximately 20% of the data.

```
quantile(propensities, 0.2)
```

```
##        20%
## 0.02968723
```

3. We first try to estimate the average causal effect for the program using inverse probability of treatment weighting (IPTW). Assess the quality of your fitted propensity model and argue

its adequacy. (You may need to refit the propensity model based on the new query in the previous subproblem)

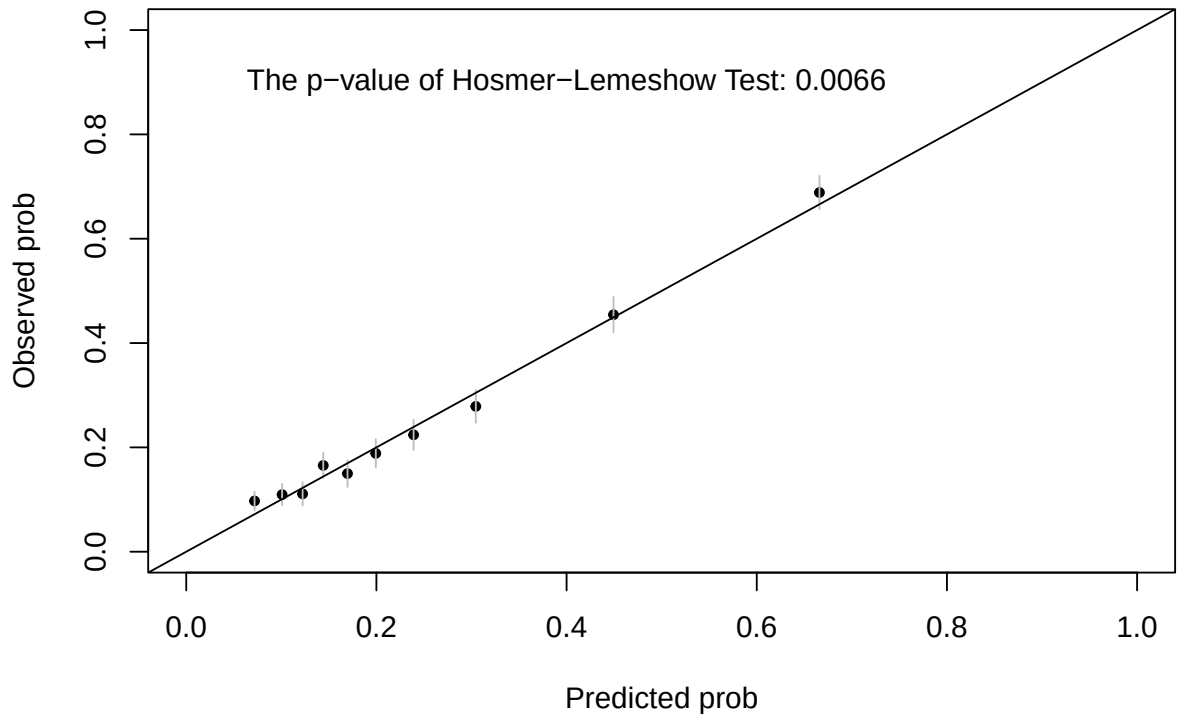
To answer the new query, we first refit the propensity model using the truncated data with only individuals who have propensities greater than or equal to 3%. The summary of the fitted model is shown below.

```
truncated = propensities >= 0.03
dat = dat[truncated, ]
fit2 = glm(prog~rcs(age, 6)+educ+rcs(sal05, 5)+race, family = "binomial", data = dat)
summary(fit2)
```

```
##
## Call:
## glm(formula = prog ~ rcs(age, 6) + educ + rcs(sal05, 5) + race,
##     family = "binomial", data = dat)
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)           -6.877e+00  1.131e+00  -6.080 1.20e-09 ***
## rcs(age, 6)age         2.105e-01  4.982e-02   4.226 2.38e-05 ***
## rcs(age, 6)age'       -2.930e+00  1.134e+00  -2.583 0.009796 **
## rcs(age, 6)age''       8.780e+00  3.612e+00   2.431 0.015062 *
## rcs(age, 6)age'''     -7.341e+00  3.221e+00  -2.279 0.022665 *
## rcs(age, 6)age''''     2.040e+00  1.048e+00   1.947 0.051509 .
## educ                   9.822e-02  9.871e-03   9.950  < 2e-16 ***
## rcs(sal05, 5)sal05    -7.138e-04  5.409e-05 -13.197  < 2e-16 ***
## rcs(sal05, 5)sal05'    1.225e-02  1.314e-03   9.320  < 2e-16 ***
## rcs(sal05, 5)sal05''  -1.462e-02  1.604e-03  -9.115  < 2e-16 ***
## rcs(sal05, 5)sal05'''  4.128e-03  5.934e-04   6.957 3.48e-12 ***
## raceHispanic           8.026e-01  1.384e-01   5.800 6.65e-09 ***
## raceOthers             3.903e-01  1.081e-01   3.612 0.000304 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 9180.5  on 8215  degrees of freedom
## Residual deviance: 7872.2  on 8203  degrees of freedom
## AIC: 7898.2
##
## Number of Fisher Scoring iterations: 4
```

To check the adequacy of the current model, we make a calibration plot and examine its calibration using Hosmer-Lemeshow test.

```
plot.calibration(dat, fit2)
```



We find that this model does not seem adequate as the p-value of the Hosmer-Lemeshow test is smaller than 0.05. As a result, we try to increase the extent of non-linearity of `age` and `sal05` in the model. Specifically, we set the restricted cubic splines of $k = 6$ for `sal05`. The refitted model is shown below.

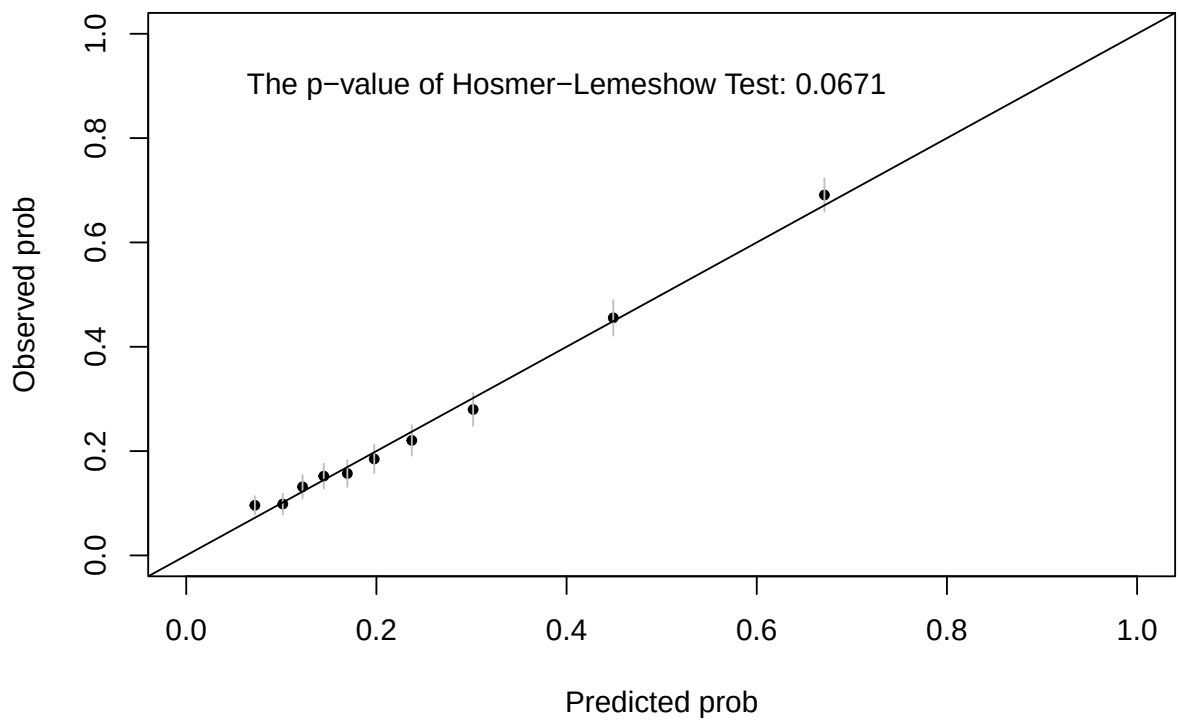
```
fit2 = glm(prog~rcs(age, 6)+educ+rcs(sal05, 6)+race, family = "binomial", data = dat)
summary(fit2)
```

```
##
## Call:
## glm(formula = prog ~ rcs(age, 6) + educ + rcs(sal05, 6) + race,
##     family = "binomial", data = dat)
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)            -6.887e+00  1.132e+00  -6.083 1.18e-09 ***
## rcs(age, 6)age          2.121e-01  4.988e-02   4.252 2.12e-05 ***
## rcs(age, 6)age'        -2.978e+00  1.136e+00  -2.623 0.008726 **
## rcs(age, 6)age''        8.930e+00  3.616e+00   2.469 0.013533 *
## rcs(age, 6)age'''      -7.462e+00  3.225e+00  -2.314 0.020689 *
## rcs(age, 6)age''''      2.059e+00  1.049e+00   1.962 0.049779 *
## educ                    9.896e-02  9.893e-03  10.003  < 2e-16 ***
## rcs(sal05, 6)sal05     -8.714e-04  7.032e-05 -12.392  < 2e-16 ***
## rcs(sal05, 6)sal05'     1.867e-02  2.201e-03   8.486  < 2e-16 ***
## rcs(sal05, 6)sal05''   -2.324e-02  2.841e-03  -8.179 2.87e-16 ***
## rcs(sal05, 6)sal05'''   6.892e-03  1.312e-03   5.253 1.49e-07 ***
```

```
## rcs(sal05, 6)sal05'''' -2.973e-03  1.559e-03  -1.907 0.056505 .
## raceHispanic            7.996e-01  1.385e-01   5.772 7.81e-09 ***
## raceOthers              3.890e-01  1.082e-01   3.596 0.000324 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 9180.5  on 8215  degrees of freedom
## Residual deviance: 7858.6  on 8202  degrees of freedom
## AIC: 7886.6
##
## Number of Fisher Scoring iterations: 4
```

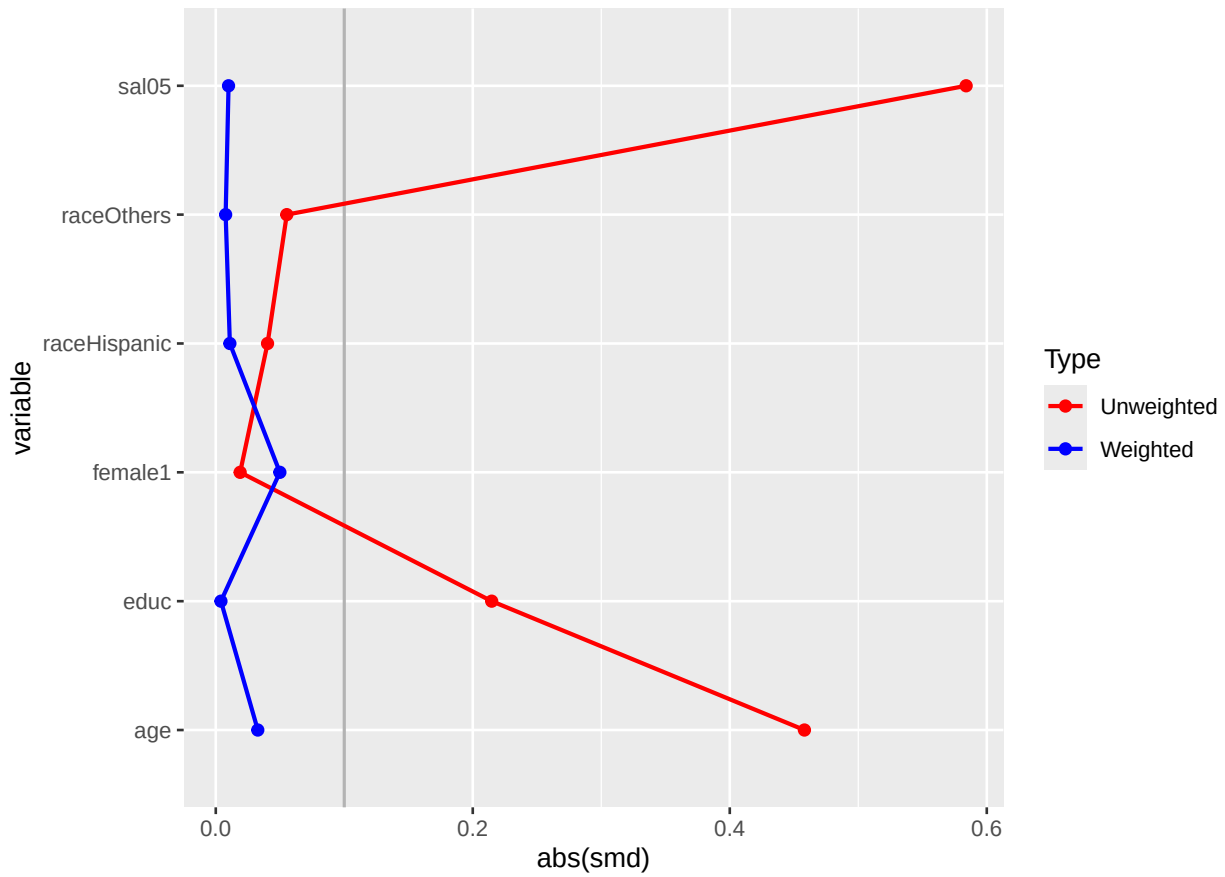
To check the adequacy of this refitted model, we make a calibration plot and examine its calibration using Hosmer-Lemeshow test.

```
plot.calibration(dat, fit2)
```



Since the p-value is greater than 0.05, we conclude that the current fitted model is well calibrated. Additionally, we should check if the covariate distribution between two treatment groups are identical after reweighting the data with weights using the inverse propensities predicted by the fitted model. In particular, we assess the standardized mean difference (SMD) of the covariates. The result is shown below.

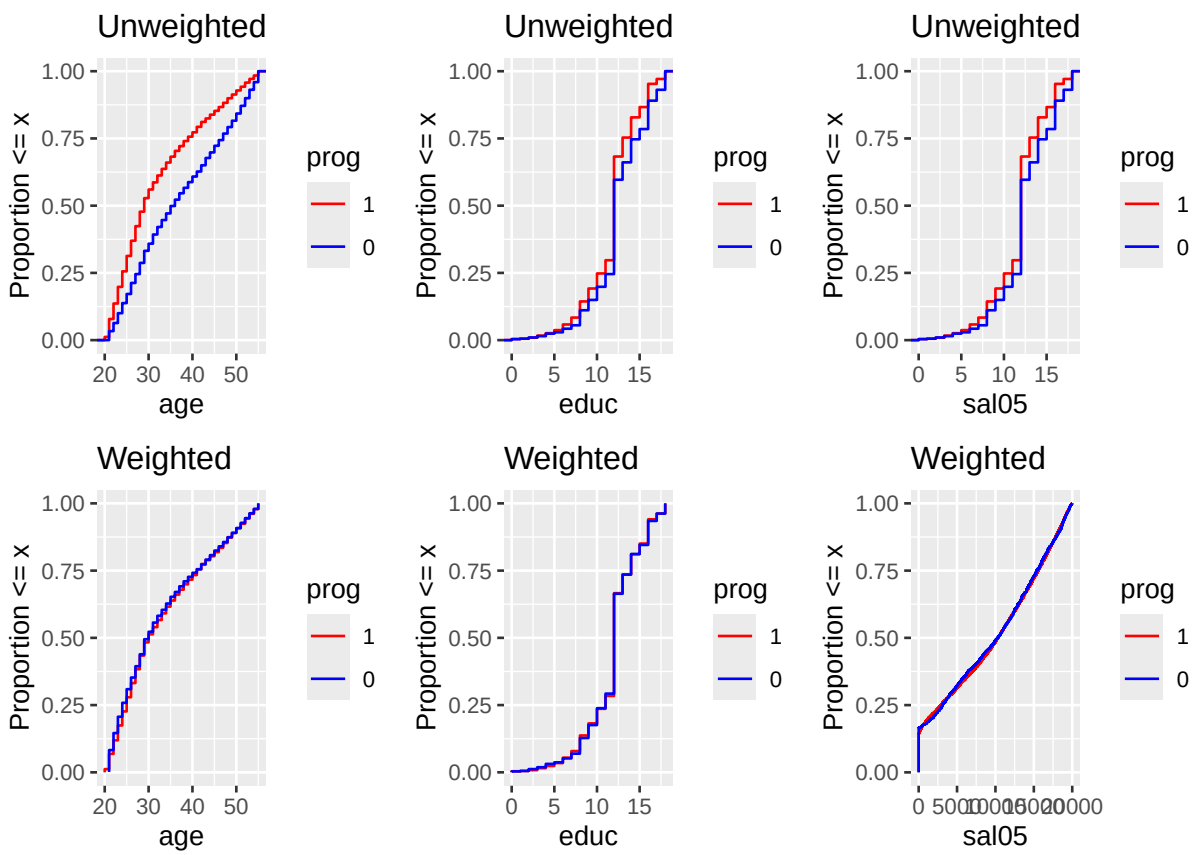
```r
propensities = predict(fit2, type = "response")
w = ifelse(dat$prog == 1, 1/propensities, -1/(1-propensities))
dat$wts = ifelse(dat$prog == 1, w, -w)

# Calculate the SMDs
smds <- tidy_smd(
  dat,
  .vars = c("age", "educ", "sal05", "female", "race"),
  .group = prog,
  .wts = wts,
  make_dummy_vars = TRUE
)

ggplot(data = smds, aes(x = abs(smd), y = variable, group = method, color = method))+
scale_color_manual("Type",
values = c("red", "blue"),
labels = c("Unweighted", "Weighted"))+
geom_love()
```



We can see the absolute value of SMDs are all less than 0.1, indicating that the current model leads to a reweighted data with covariate balance properties. To be more confident in the covariate balance, we may also compare the estimated cumulative distribution function of each of the continuous variables: `age`, `educ`, `sal05`, between the two treatment groups. The results are shown below.

```r
# Plot eCDFs of the weight variable for comparison
p1 <- ggplot(dat, aes(x = age, color = factor(prog))) +
geom_ecdf() +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("age") +
ylab("Proportion <= x") +
ggtitle("Unweighted")

p2 <- ggplot(dat, aes(x = age, color = factor(prog))) +
geom_ecdf(aes(weights = wts)) +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("age") +
ylab("Proportion <= x") +
ggtitle("Weighted")

p3 <- ggplot(dat, aes(x = educ, color = factor(prog))) +
geom_ecdf() +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("educ") +
ylab("Proportion <= x") +
ggtitle("Unweighted")

p4 <- ggplot(dat, aes(x = educ, color = factor(prog))) +
geom_ecdf(aes(weights = wts)) +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("educ") +
ylab("Proportion <= x") +
ggtitle("Weighted")

p5 <- ggplot(dat, aes(x = educ, color = factor(prog))) +
geom_ecdf() +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("sal05") +
```

```
ylab("Proportion <= x") +
ggtitle("Unweighted")

p6 <- ggplot(dat, aes(x = sal05, color = factor(prog))) +
geom_ecdf(aes(weights = wts)) +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("sal05") +
ylab("Proportion <= x") +
ggtitle("Weighted")

layout <- (p1 | p3 | p5) /
          (p2 | p4 | p6)

layout
```



We can see the empirical cumulative distribution functions for each continuous covariate in the two treatment groups align closely after the data is reweighted. This provides further evidence that the covariate balance has been achieved using the current model. Overall, we conclude that the refitted model is sufficiently adequate. Finally, we should verify whether the minimum value of the propensities is now approximately 0.03.

```
min(propensities)
```

```
## [1] 0.03469657
```

The minimum value of propensities is, as expected, greater than 0.03. Now, We can compute the IPTW estimator as follows.

```
iptw = mean(w*dat$sal07)
```

Using the IPTW estimator, we find that the average causal effect is 561.98.

4. Use the IPTW approach to estimate the average causal effect of the program. Argue if this effect is statistically significant using hypothesis tests.

We may check the significance of the IPTW estimate of the average causal effect by computing the non-parametric 95% bootstrapped confidence interval. Denote τ as the average causal effect. The null and alternative hypotheses are as follows.

$$\begin{cases} H_0 : \tau = 0 \\ H_1 : \tau \neq 0. \end{cases}$$

To compute the non-parametric 95% confidence interval, we carry out the following procedure:

step 1. Obtain the bth bootstrapped sample with the same sample size as the original data through simple random sampling with replacement of the original data.

step 2. Fit the propensity model which we fitted in question 3, and compute the IPTW estimator of τ, denoted by $\widehat{\tau}_{iptw}^{(b)}$.

step 3. Repeat step 1 and step 2 for $b = 1, \ldots, 500$, and compute the 0.025 and 0.975 quantiles of the boostrapped statistics $\widehat{\tau}_{iptw}^{(1)}, \ldots, \widehat{\tau}_{iptw}^{(500)}$.

```
get_iptw = function(dat){
  fit = glm(prog~rcs(age, 6)+educ+rcs(sal05, 6)+race, family = "binomial", data = dat)
  propensities = predict(fit, type = "response")
  w = ifelse(dat$prog == 1, dat$prog/propensities, -(1-dat$prog)/(1-propensities))
  iptw = mean(w*dat$sal07)
  return(iptw)
}
```

```
B = 500
boot_iptw = rep(NA, B)
for(b in 1:B){
  boot_sample = dat[sample(1:nrow(dat), replace = T), ]
  boot_iptw[b] = get_iptw(boot_sample)
}
saveRDS(boot_iptw, "boot_iptw.rds")
```

```
boot_iptw = readRDS("boot_iptw.rds")
iptw.ci = round(quantile(boot_iptw, c(0.025, 0.975)), 2)
```

The 95% confidence interval is [353.09, 760.45]. Since the confidence interval does not contain 0, we conclude that the IPTW estimate of the average causal effect is significant at 0.05 significance level.

5. Alternatively, use the standardization (outcome regression) approach to estimate the average causal effect. Argue if this effect is statistically significant using hypothesis tests. Compare the estimate with the estimate from the IPTW approach.

To apply the standardization approach, we need to fit two separate outcome regression models: one for individuals who participated in the program and another for those who did not. Specifically, we fit the linear regression models as follows.

$$\mu_1(L) = E(Y \mid A = 1) = \alpha_0 + \alpha_1^T b_1(L),$$
$$\mu_0(L) = E(Y \mid A = 0) = \gamma_0 + \gamma_1^T b_0(L),$$

where $b_1(L)$ and $b_0(L)$ are basis functions including restricted cubic splines of $k = 4$ for `age`, `educ` and `sal05`, and the dummy variables for `race` and `female`. The summary tables of the two fitted models are shown below.

```
outcome1 = lm(sal07~rcs(age, 4)+rcs(educ, 4)+rcs(sal05, 4)+race+female, data = dat, subset = prog == 1)
outcome0 = lm(sal07~rcs(age, 4)+rcs(educ, 4)+rcs(sal05, 4)+race+female, data = dat, subset = prog == 0)
summary(outcome1)
```

```
##
## Call:
## lm(formula = sal07 ~ rcs(age, 4) + rcs(educ, 4) + rcs(sal05,
##     4) + race + female, data = dat, subset = prog == 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -25536.3  -1493.5   -117.9   1264.9  20723.9
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -1.417e+04  1.471e+03  -9.633  < 2e-16 ***
## rcs(age, 4)age        1.797e+02  5.473e+01   3.283  0.00105 **
## rcs(age, 4)age'      -8.226e+02  3.245e+02  -2.535  0.01132 *
## rcs(age, 4)age''      1.653e+03  6.096e+02   2.711  0.00676 **
## rcs(educ, 4)educ      5.746e+02  4.514e+01  12.729  < 2e-16 ***
## rcs(educ, 4)educ'     8.710e+01  4.663e+01   1.868  0.06196 .
## rcs(sal05, 4)sal05    1.518e+00  8.743e-02  17.367  < 2e-16 ***
## rcs(sal05, 4)sal05'  -3.838e+00  1.584e+00  -2.423  0.01549 *
## rcs(sal05, 4)sal05''  4.415e+00  1.830e+00   2.413  0.01589 *
## raceHispanic          2.378e+03  3.234e+02   7.354 2.78e-13 ***
## raceOthers            1.055e+03  2.572e+02   4.104 4.23e-05 ***
## female1               6.716e+01  1.270e+02   0.529  0.59702
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2855 on 2015 degrees of freedom
## Multiple R-squared:  0.9135, Adjusted R-squared:  0.913
## F-statistic:  1933 on 11 and 2015 DF,  p-value: < 2.2e-16
```

```
summary(outcome0)
```

```
##
## Call:
```

```
## lm(formula = sal07 ~ rcs(age, 4) + rcs(educ, 4) + rcs(sal05,
##     4) + race + female, data = dat, subset = prog == 0)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22211.9  -1199.2    -38.5   1058.3  15510.8
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -1.258e+04  5.163e+02 -24.367  < 2e-16 ***
## rcs(age, 4)age        8.259e+01  1.925e+01   4.291 1.81e-05 ***
## rcs(age, 4)age'       2.092e+01  1.260e+02   0.166 0.868110
## rcs(age, 4)age''      1.602e+01  2.436e+02   0.066 0.947568
## rcs(educ, 4)educ      5.912e+02  1.936e+01  30.531  < 2e-16 ***
## rcs(educ, 4)educ'     1.645e+02  2.227e+01   7.387 1.70e-13 ***
## rcs(sal05, 4)sal05    1.301e+00  4.160e-02  31.273  < 2e-16 ***
## rcs(sal05, 4)sal05'  -2.557e+00  6.654e-01  -3.843 0.000123 ***
## rcs(sal05, 4)sal05''  2.956e+00  7.641e-01   3.868 0.000111 ***
## raceHispanic          3.058e+03  1.320e+02  23.163  < 2e-16 ***
## raceOthers            2.064e+03  9.423e+01  21.902  < 2e-16 ***
## female1               9.331e-01  5.550e+01   0.017 0.986586
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2182 on 6177 degrees of freedom
## Multiple R-squared:  0.9223, Adjusted R-squared:  0.9222
## F-statistic:  6667 on 11 and 6177 DF,  p-value: < 2.2e-16
```

As seen in the summary tables, we remove `female` from the models as it appears not significant according to the Wald's test. The refitted model is shown below.

```
outcome1 = lm(sal07~rcs(age, 4)+rcs(educ, 4)+rcs(sal05, 4)+race, data = dat, subset = prog == 1)
outcome0 = lm(sal07~rcs(age, 4)+rcs(educ, 4)+rcs(sal05, 4)+race, data = dat, subset = prog == 0)
summary(outcome1)
```

```
##
## Call:
## lm(formula = sal07 ~ rcs(age, 4) + rcs(educ, 4) + rcs(sal05,
##     4) + race, data = dat, subset = prog == 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -25501.9  -1503.1   -107.9   1258.4  20758.1
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -1.413e+04  1.469e+03  -9.621  < 2e-16 ***
## rcs(age, 4)age        1.796e+02  5.472e+01   3.282  0.00105 **
## rcs(age, 4)age'      -8.215e+02  3.244e+02  -2.532  0.01142 *
## rcs(age, 4)age''      1.650e+03  6.094e+02   2.708  0.00683 **
## rcs(educ, 4)educ      5.748e+02  4.513e+01  12.737  < 2e-16 ***
## rcs(educ, 4)educ'     8.690e+01  4.662e+01   1.864  0.06248 .
## rcs(sal05, 4)sal05    1.518e+00  8.741e-02  17.364  < 2e-16 ***
```

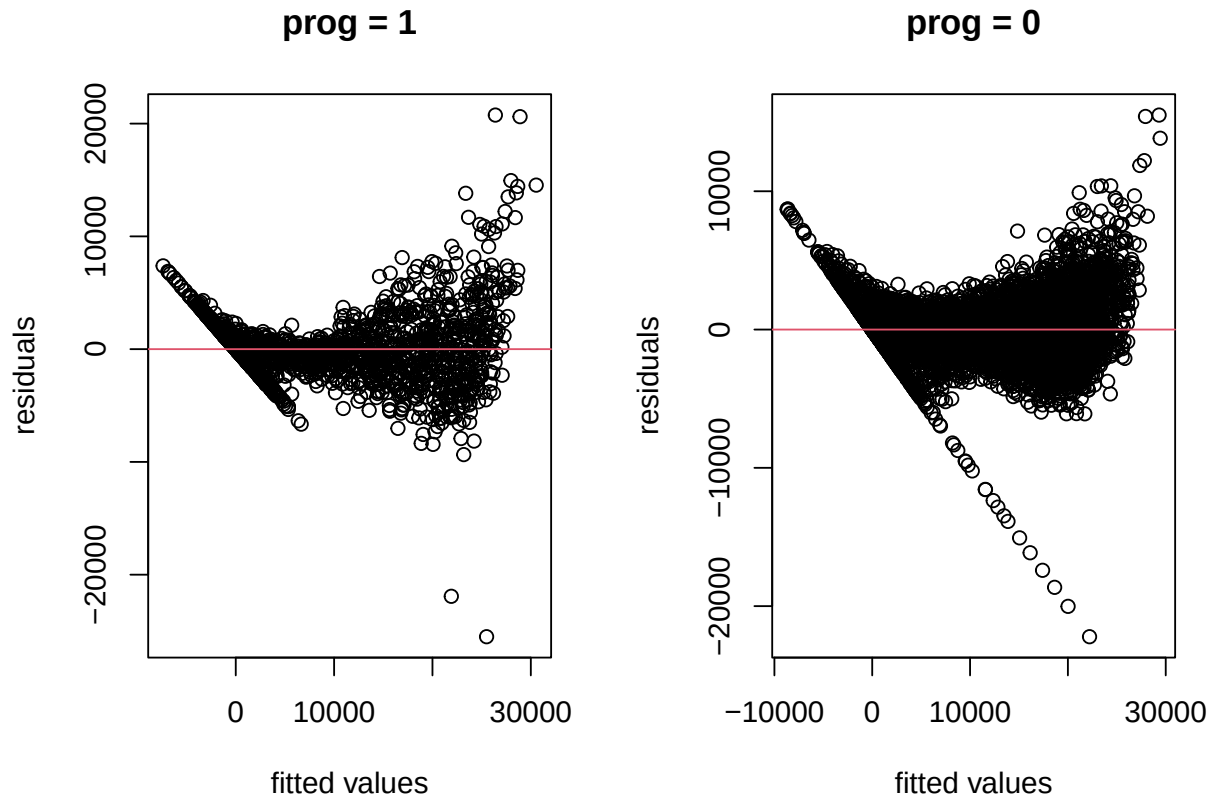
```
## rcs(sal05, 4)sal05'  -3.829e+00  1.584e+00  -2.418  0.01570 *
## rcs(sal05, 4)sal05''  4.405e+00  1.829e+00   2.408  0.01611 *
## raceHispanic          2.375e+03  3.233e+02   7.347 2.93e-13 ***
## raceOthers            1.054e+03  2.571e+02   4.100 4.30e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2855 on 2016 degrees of freedom
## Multiple R-squared:  0.9134, Adjusted R-squared:  0.913
## F-statistic:  2127 on 10 and 2016 DF,  p-value: < 2.2e-16
```

```r
summary(outcome0)
```

```
##
## Call:
## lm(formula = sal07 ~ rcs(age, 4) + rcs(educ, 4) + rcs(sal05,
##     4) + race, data = dat, subset = prog == 0)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22212.4  -1199.5    -38.1   1058.7  15511.3
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -1.258e+04  5.156e+02 -24.399  < 2e-16 ***
## rcs(age, 4)age        8.259e+01  1.924e+01   4.291 1.80e-05 ***
## rcs(age, 4)age'       2.091e+01  1.260e+02   0.166 0.868160
## rcs(age, 4)age''      1.604e+01  2.436e+02   0.066 0.947499
## rcs(educ, 4)educ      5.912e+02  1.936e+01  30.534  < 2e-16 ***
## rcs(educ, 4)educ'     1.645e+02  2.226e+01   7.389 1.68e-13 ***
## rcs(sal05, 4)sal05    1.301e+00  4.159e-02  31.277  < 2e-16 ***
## rcs(sal05, 4)sal05'  -2.557e+00  6.653e-01  -3.843 0.000123 ***
## rcs(sal05, 4)sal05''  2.956e+00  7.640e-01   3.869 0.000111 ***
## raceHispanic          3.058e+03  1.320e+02  23.166  < 2e-16 ***
## raceOthers            2.064e+03  9.423e+01  21.904  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2181 on 6178 degrees of freedom
## Multiple R-squared:  0.9223, Adjusted R-squared:  0.9222
## F-statistic:  7335 on 10 and 6178 DF,  p-value: < 2.2e-16
```

To check the adequacy of the both fitted models, we make the residual plots for each of them.

```r
par(mfrow = c(1, 2))
plot(residuals(outcome1)~fitted(outcome1), xlab = "fitted values", ylab = "residuals",
     main = "prog = 1")
abline(h = 0, col = 2)
plot(residuals(outcome0)~fitted(outcome0), xlab = "fitted values", ylab = "residuals",
     main = "prog = 0")
abline(h = 0, col = 2)
```



Both models are clearly unsatisfactory. To improve their goodness-of-fit, we adjust the non-linearity of `age`, `educ`, and `sal05` by adjusting the parameter k in the `rcs` function. This process continues until further increments in k result in insignificant non-linear terms in the model. Ultimately, this approach leads to the following settings of basis functions for the outcome regression models:

1. Basis functions of $\mu_1(L)$:

- Restricted cubic splines with $k = 4$ for `age`.
- Restricted cubic splines with $k = 6$ for `educ`.
- Restricted cubic splines with $k = 4$ for `sal05`.
- Dummy variables of `race`‘.

2. Basis functions of $\mu_0(L)$:

- Restricted cubic splines with $k = 3$ for `age`.
- Restricted cubic splines with $k = 8$ for `educ`.
- Restricted cubic splines with $k = 5$ for `sal05`.
- Dummy variables of `race`.

```
outcome1 = lm(sal07~rcs(age, 4)+rcs(educ, 6)+rcs(sal05, 4)+race, data = dat, subset = prog == 1)
outcome0 = lm(sal07~rcs(age, 3)+rcs(educ, 8)+rcs(sal05, 5)+race, data = dat, subset = prog == 0)
summary(outcome1)
```

```
##
```

```
## Call:
## lm(formula = sal07 ~ rcs(age, 4) + rcs(educ, 6) + rcs(sal05,
##     4) + race, data = dat, subset = prog == 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -25420.7  -1499.3   -219.2   1161.4  20952.4
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -1.271e+04  1.502e+03  -8.465  < 2e-16 ***
## rcs(age, 4)age        1.863e+02  5.457e+01   3.413 0.000654 ***
## rcs(age, 4)age'      -8.472e+02  3.231e+02  -2.622 0.008807 **
## rcs(age, 4)age''      1.696e+03  6.068e+02   2.795 0.005240 **
## rcs(educ, 6)educ      3.004e+02  7.603e+01   3.951 8.04e-05 ***
## rcs(educ, 6)educ'     1.357e+03  3.152e+02   4.304 1.76e-05 ***
## rcs(educ, 6)educ''   -6.408e+03  2.075e+03  -3.088 0.002044 **
## rcs(educ, 6)educ'''   7.182e+03  3.813e+03   1.884 0.059744 .
## rcs(sal05, 4)sal05    1.510e+00  8.702e-02  17.354  < 2e-16 ***
## rcs(sal05, 4)sal05'  -3.634e+00  1.577e+00  -2.304 0.021319 *
## rcs(sal05, 4)sal05''  4.177e+00  1.821e+00   2.293 0.021928 *
## raceHispanic          2.274e+03  3.225e+02   7.052 2.42e-12 ***
## raceOthers            1.027e+03  2.560e+02   4.010 6.29e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2842 on 2014 degrees of freedom
## Multiple R-squared:  0.9143, Adjusted R-squared:  0.9138
## F-statistic:  1791 on 12 and 2014 DF,  p-value: < 2.2e-16
```

```
summary(outcome0)
```

```
##
## Call:
## lm(formula = sal07 ~ rcs(age, 3) + rcs(educ, 8) + rcs(sal05,
##     5) + race, data = dat, subset = prog == 0)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22445.9  -1210.8    -52.3   1061.6  15603.9
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -1.125e+04  3.810e+02 -29.518  < 2e-16 ***
## rcs(age, 3)age        8.400e+01  9.562e+00   8.785  < 2e-16 ***
## rcs(age, 3)age'       4.005e+01  1.812e+01   2.210  0.02714 *
## rcs(educ, 8)educ      3.195e+02  4.041e+01   7.907 3.10e-15 ***
## rcs(educ, 8)educ'     1.201e+03  1.566e+02   7.667 2.03e-14 ***
## rcs(educ, 8)educ''   -6.081e+03  1.233e+03  -4.933 8.33e-07 ***
## rcs(educ, 8)educ'''   2.432e+04  8.182e+03   2.973  0.00296 **
## rcs(educ, 8)educ''''  -2.718e+04  1.063e+04  -2.556  0.01060 *
## rcs(sal05, 5)sal05    1.205e+00  5.905e-02  20.409  < 2e-16 ***
## rcs(sal05, 5)sal05'   2.389e-01  1.324e+00   0.181  0.85675
## rcs(sal05, 5)sal05''  -5.054e-01  1.606e+00  -0.315  0.75298
```

```
## rcs(sal05, 5)sal05'''  1.240e+00  5.570e-01   2.226  0.02605 *
## raceHispanic           3.049e+03  1.313e+02  23.217  < 2e-16 ***
## raceOthers             2.080e+03  9.376e+01  22.189  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2170 on 6175 degrees of freedom
## Multiple R-squared:  0.9232, Adjusted R-squared:  0.923
## F-statistic:  5706 on 13 and 6175 DF,  p-value: < 2.2e-16
```

Then, we calculate the standardization estimate of the average causal effect as follows.

```
mu1 = mean(predict(outcome1, dat))
mu0 = mean(predict(outcome0, dat))
stand = mu1 - mu0
```

Using the standardization estimator, we find that the average causal effect is 584.39.

We may check the significance of the standardization estimate of the average causal effect by computing the non-parametric 95% bootstrapped confidence interval. The null and alternative hypotheses are as follows.

$$\begin{cases} H_0 : \tau = 0 \\ H_1 : \tau \neq 0. \end{cases}$$

To compute the non-parametric 95% confidence interval, we carry out the following procedure:

step 1. Obtain the bth bootstrapped sample with the same sample size as the original data through simple random sampling with replacement of the original data.

step 2. Fit the two separate outcome regression models: one for individuals who participated in the program and another for those who did not using the models which we just fitted, and compute the standardization estimator of τ, denoted by $\hat{\tau}^{(b)}_{stand}$.

step 3. Repeat step 1 and step 2 for $b = 1, \ldots, 500$, and compute the 0.025 and 0.975 quantiles of the boostrapped statistics $\hat{\tau}^{(1)}_{stand}, \ldots, \hat{\tau}^{(500)}_{stand}$.

```
get_stand = function(dat){
  outcome1 = lm(sal07~rcs(age, 4)+rcs(educ, 6)+rcs(sal05, 4)+race, data = dat, subset = prog == 1)
  outcome0 = lm(sal07~rcs(age, 3)+rcs(educ, 8)+rcs(sal05, 5)+race, data = dat, subset = prog == 0)
  mu1 = mean(predict(outcome1, dat))
  mu0 = mean(predict(outcome0, dat))
  stand = mu1 - mu0
  return(stand)
}
```

```
B = 500
boot_stand = rep(NA, B)
for(b in 1:B){
  boot_sample = dat[sample(1:nrow(dat), replace = T), ]
  boot_stand[b] = get_stand(boot_sample)
}
saveRDS(boot_stand, "boot_stand.rds")
```

```
boot_stand = readRDS("boot_stand.rds")
stand.ci = round(quantile(boot_stand, c(0.025, 0.975)), 2)
```

The 95% confidence interval is [421.42, 751.65]. Since the confidence interval does not contain 0, we conclude that the standardization estimate of the average causal effect is significant at 0.05 significance level. Interestingly, this conclusion is consistent with the results obtained using the IPTW estimator. Moreover, both approaches result in similar estimates and confidence intervals.

6. The city administrator would also like to know if the program should be focused on specific populations to yield maximum benefit. He would like to start by assessing if the program benefits more to citizens that have attended college or university (i.e. have more than 12 years of education). Answer the administrator's query with hypothesis tests.

For simplicity, we creat a new variable Z with

$$\begin{cases} Z = 1, & \text{if have more than 12 years of education} \\ Z = 0, & \text{if have at most 12 years of education.} \end{cases}$$

Define $\tau_1 = E(Y(1) - Y(1) \mid Z = 1)$ and $\tau_0 = E(Y(1) - Y(1) \mid Z = 0)$. To answer if the program benefits more to citizens that have attended college or university, we should estimate the difference $\tau_1 - \tau_0$. By question 5, since we found that our IPTW and standardization approaches generally lead to the same results for this data, we conduct only the IPTW analysis to answer the question here.

```
w = ifelse(dat$prog[dat$educ > 12] == 1, 1/propensities[dat$educ > 12],
           -1/(1-propensities[dat$educ > 12]))
iptw1 = mean(w*dat$sal07[dat$educ > 12])
w = ifelse(dat$prog[dat$educ <= 12] == 1, 1/propensities[dat$educ <= 12],
           -1/(1-propensities[dat$educ <= 12]))
iptw0 = mean(w*dat$sal07[dat$educ <= 12])
iptw_diff_educ = iptw1 - iptw0
```

Denote the IPTW estimates for τ_1 and τ_2 as $\hat{\tau_1}$ and $\hat{\tau}_2$ respectively. We obtain $\hat{\tau}_1 = 1525.22$ and $\hat{\tau}_0 = 69.12$, and hence the difference of interest is 1456.1. To check the significance of the difference, we should perform a hypothesis test with the following hypotheses:

$$\begin{cases} H_0 : \tau_1 = \tau_0 & (\tau_1 - \tau_0 = 0) \\ H_1 : \tau_1 > \tau_0 & (\tau_1 - \tau_0 > 0). \end{cases}$$

We may perform this test by computing the non-parametric 95% bootstrapped one-sided confidence interval of $\tau_1 - \tau_0$. Specifically, we carry out the following procedure:

step 1. Obtain the bth bootstrapped sample with the same sample size as the original data through simple random sampling with replacement of the original data.

step 2. Fit the propensity model which we fitted in question 3, and compute the IPTW estimator of τ_1 and τ_0, denoted as $\hat{\tau}_1^{(b)}$ and $\hat{\tau}_0^{(b)}$ respectively. Then compute their difference $\hat{\tau}_1^{(b)} - \hat{\tau}_0^{(b)}$.

step 3. Repeat step 1 and step 2 for $b = 1, \ldots, 500$, and compute the 0.05 quantile of the boostrapped statistics $\hat{\tau}_1^{(1)} - \hat{\tau}_0^{(1)}, \ldots, \hat{\tau}_1^{(500)} - \hat{\tau}_0^{(500)}$.

```
boot_iptw_diff_educ = readRDS("boot_iptw_diff_educ.rds")
iptw_diff_educ.lower = round(quantile(boot_iptw_diff_educ, 0.05), 2)
```

The 95% one-sided confidence interval for $\tau_1 - \tau_0$ is $[76.35, \infty]$. Since this confidence interval does not contain 0, we conclude that the program benefits more to citizens that have attended college or university by increasing their salary by 1456.1 dollars more than those that have never attended college or university.

7. Provide the estimates of conditional average causal effects for citizens with more than 12 years and at most 12 years of education. Use hypothesis tests to determine if the causal effects are statistically significant within each subpopulation.

Similar to question 6, we only report our results based on the IPTW approach. In question 6, we obtained that $\hat{\tau}_1 = 1525.22$ and $\hat{\tau}_0 = 69.12$. To check if $\hat{\tau}_1$ and $\hat{\tau}_0$ are statistically significant within each subpopulation, we should perform hypothesis tests:

$$\begin{cases} H_0 : \tau_1 = 0 \\ H_1 : \tau_1 \neq 0 \end{cases}$$

and

$$\begin{cases} H_0 : \tau_0 = 0 \\ H_1 : \tau_0 \neq 0 \end{cases}$$

respectively.

We may perform this test by computing the non-parametric 95% bootstrapped confidence interval of τ_j for $j = 0, 1$. Specifically, we carry out the following procedure for each j:

step 1. Obtain the bth bootstrapped sample with the same sample size as the original data through simple random sampling with replacement of the original data.

step 2. Fit the propensity model which we fitted in question 3, and compute the IPTW estimator of τ_j, denoted by $\hat{\tau}_j^{(b)}$.

step 3. Repeat step 1 and step 2 for $b = 1, \ldots, 500$, and compute the 0.025 and 0.975 quantiles of the boostrapped statistics $\hat{\tau}_j^{(1)}, \ldots, \hat{\tau}_j^{(500)}$.

```
get_iptw1 = function(dat){
  fit = glm(prog~rcs(age, 6)+educ+rcs(sal05, 6)+race, family = "binomial", data = dat)
  propensities = predict(fit, type = "response")
  w = ifelse(dat$prog[dat$educ > 12] == 1, 1/propensities[dat$educ > 12],
             -1/(1-propensities[dat$educ > 12]))
  iptw1 = mean(w*dat$sal07[dat$educ > 12])
  return(iptw1 - iptw0)
}

get_iptw0 = function(dat){
  fit = glm(prog~rcs(age, 6)+educ+rcs(sal05, 6)+race, family = "binomial", data = dat)
  propensities = predict(fit, type = "response")
  w = ifelse(dat$prog[dat$educ <= 12] == 1, 1/propensities[dat$educ <= 12],
             -1/(1-propensities[dat$educ <= 12]))
  iptw0 = mean(w*dat$sal07[dat$educ <= 12])
  return(iptw0)
}
```

```
B = 500
boot_iptw1 = boot_iptw0 = rep(NA, B)
for(b in 1:B){
  boot_sample = dat[sample(1:nrow(dat), replace = T), ]
  boot_iptw1[b] = get_iptw1(boot_sample)
```

```
  boot_iptw0[b] = get_iptw0(boot_sample)
}
saveRDS(boot_iptw1, "boot_iptw1.rds")
saveRDS(boot_iptw0, "boot_iptw0.rds")
```

```
boot_iptw1 = readRDS("boot_iptw1.rds")
iptw1.ci = round(quantile(boot_iptw1, c(0.025, 0.975)), 2)
boot_iptw0 = readRDS("boot_iptw0.rds")
iptw0.ci = round(quantile(boot_iptw0, c(0.025, 0.975)), 2)
```

The 95% confidence interval for τ_1 is [376.74, 2671.06]. Since this confidence interval does not contain 0, we conclude that the IPTW estimate of the average causal effect for citizens that have attended college or university is significant at 0.05 significance level. In contrast, the 95% confidence interval for τ_0 is [-490.02, 642.22]. Since this confidence interval contains 0, we conclude that the IPTW estimate of the average causal effect for citizens that have never attended college or university is NOT significant at 0.05 significance level.

3 Source Code

```
knitr::opts_chunk$set(echo = TRUE, fig.width = 7, fig.height = 5)
library(kableExtra)
library(tidyverse)
library(dplyr)
library(knitr)
library(rms)
library(pROC)
library(lme4)
library(lmeresampler)
library(pbkrtest)
library(geepack)
library(halfmoon)
library(patchwork)
dat = read.csv("salary.csv")
dat$female = as.factor(dat$female)
dat$race = as.factor(dat$race)
fit1 = glm(prog~rcs(age, 4)+rcs(educ, 4)+rcs(sal05, 4)+female+race, family = "binomial", data = dat)
summary(fit1)
fit1 = glm(prog~rcs(age, 4)+educ+rcs(sal05, 4)+race, family = "binomial", data = dat)
summary(fit1)
plot.calibration = function(dat, fit){
  propensities = predict(fit, type = "response")
  dat$fitted = propensities
  dat$group = cut(dat$fitted,
  quantile(dat$fitted, (0:10)/10),
  include.lowest = T)
  hl.dat = dat %>%
  group_by(group) %>%
  dplyr::summarize(sum_prob = sum(fitted), sum_non_prob = sum(1-fitted),
  sum_event = sum(prog == 1), sum_non_event = sum(prog == 0),
  mean_prob = mean(fitted), mean_event = mean(prog == 1),
  n = n()) %>%
  mutate(se = sqrt(mean_prob*(1-mean_prob)/n),
  lower = pmax(0, mean_event-1.96*se),
  upper = pmin(1, mean_event+1.96*se))
  plot(hl.dat$mean_prob, hl.dat$mean_event, xlim = c(0, 1), ylim = c(0, 1),
  xlab = "Predicted prob", ylab = "Observed prob", pch = 20)
  for(i in 1:nrow(hl.dat)){
  points(hl.dat$mean_prob[c(i, i)], c(hl.dat$lower[i], hl.dat$upper[i]),
  col = "grey", type = "l")
  }
  abline(a = 0, b = 1)

  hl = sum((hl.dat$sum_event - hl.dat$sum_prob)^2/hl.dat$sum_prob) +
  sum((hl.dat$sum_non_event - hl.dat$sum_non_prob)^2/hl.dat$sum_non_prob)
  p.value = 1-pchisq(hl, 8)

  text(0.4, 0.9, paste("The p-value of Hosmer-Lemeshow Test:", round(p.value, 4)))
}
plot.calibration(dat, fit1)
fit1 = glm(prog~rcs(age, 6)+educ+rcs(sal05, 5)+race, family = "binomial", data = dat)
```

```r
summary(fit1)
plot.calibration(dat, fit1)
propensities = predict(fit1, type = "response")
boxplot(propensities)
c(min(propensities), max(propensities))
quantile(propensities, 0.2)
truncated = propensities >= 0.03
dat = dat[truncated, ]
fit2 = glm(prog~rcs(age, 6)+educ+rcs(sal05, 5)+race, family = "binomial", data = dat)
summary(fit2)
plot.calibration(dat, fit2)
fit2 = glm(prog~rcs(age, 6)+educ+rcs(sal05, 6)+race, family = "binomial", data = dat)
summary(fit2)
plot.calibration(dat, fit2)
propensities = predict(fit2, type = "response")
w = ifelse(dat$prog == 1, 1/propensities, -1/(1-propensities))
dat$wts = ifelse(dat$prog == 1, w, -w)

# Calculate the SMDs
smds <- tidy_smd(
  dat,
  .vars = c("age", "educ", "sal05", "female", "race"),
  .group = prog,
  .wts = wts,
  make_dummy_vars = TRUE
)

ggplot(data = smds, aes(x = abs(smd), y = variable, group = method, color = method))+
scale_color_manual("Type",
values = c("red", "blue"),
labels = c("Unweighted", "Weighted"))+
geom_love()
# Plot eCDFs of the weight variable for comparison
p1 <- ggplot(dat, aes(x = age, color = factor(prog))) +
geom_ecdf() +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("age") +
ylab("Proportion <= x") +
ggtitle("Unweighted")

p2 <- ggplot(dat, aes(x = age, color = factor(prog))) +
geom_ecdf(aes(weights = wts)) +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("age") +
ylab("Proportion <= x") +
```

```
ggtitle("Weighted")

p3 <- ggplot(dat, aes(x = educ, color = factor(prog))) +
geom_ecdf() +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("educ") +
ylab("Proportion <= x") +
ggtitle("Unweighted")

p4 <- ggplot(dat, aes(x = educ, color = factor(prog))) +
geom_ecdf(aes(weights = wts)) +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("educ") +
ylab("Proportion <= x") +
ggtitle("Weighted")

p5 <- ggplot(dat, aes(x = educ, color = factor(prog))) +
geom_ecdf() +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("sal05") +
ylab("Proportion <= x") +
ggtitle("Unweighted")

p6 <- ggplot(dat, aes(x = sal05, color = factor(prog))) +
geom_ecdf(aes(weights = wts)) +
scale_color_manual(
"prog",
values = c("red", "blue"),
labels = c("1", "0")
) +
xlab("sal05") +
ylab("Proportion <= x") +
ggtitle("Weighted")

layout <- (p1 | p3 | p5) /
          (p2 | p4 | p6)

layout
min(propensities)
iptw = mean(w*dat$sal07)
get_iptw = function(dat){
```

```r
  fit = glm(prog~rcs(age, 6)+educ+rcs(sal05, 6)+race, family = "binomial", data = dat)
  propensities = predict(fit, type = "response")
  w = ifelse(dat$prog == 1, dat$prog/propensities, -(1-dat$prog)/(1-propensities))
  iptw = mean(w*dat$sal07)
  return(iptw)
}
B = 500
boot_iptw = rep(NA, B)
for(b in 1:B){
  boot_sample = dat[sample(1:nrow(dat), replace = T), ]
  boot_iptw[b] = get_iptw(boot_sample)
}
saveRDS(boot_iptw, "boot_iptw.rds")
boot_iptw = readRDS("boot_iptw.rds")
iptw.ci = round(quantile(boot_iptw, c(0.025, 0.975)), 2)
outcome1 = lm(sal07~rcs(age, 4)+rcs(educ, 4)+rcs(sal05, 4)+race+female, data = dat, subset = prog == 1)
outcome0 = lm(sal07~rcs(age, 4)+rcs(educ, 4)+rcs(sal05, 4)+race+female, data = dat, subset = prog == 0)
summary(outcome1)
summary(outcome0)
outcome1 = lm(sal07~rcs(age, 4)+rcs(educ, 4)+rcs(sal05, 4)+race, data = dat, subset = prog == 1)
outcome0 = lm(sal07~rcs(age, 4)+rcs(educ, 4)+rcs(sal05, 4)+race, data = dat, subset = prog == 0)
summary(outcome1)
summary(outcome0)
par(mfrow = c(1, 2))
plot(residuals(outcome1)~fitted(outcome1), xlab = "fitted values", ylab = "residuals",
     main = "prog = 1")
abline(h = 0, col = 2)
plot(residuals(outcome0)~fitted(outcome0), xlab = "fitted values", ylab = "residuals",
     main = "prog = 0")
abline(h = 0, col = 2)
outcome1 = lm(sal07~rcs(age, 4)+rcs(educ, 6)+rcs(sal05, 4)+race, data = dat, subset = prog == 1)
outcome0 = lm(sal07~rcs(age, 3)+rcs(educ, 8)+rcs(sal05, 5)+race, data = dat, subset = prog == 0)
summary(outcome1)
summary(outcome0)
mu1 = mean(predict(outcome1, dat))
mu0 = mean(predict(outcome0, dat))
stand = mu1 - mu0
get_stand = function(dat){
  outcome1 = lm(sal07~rcs(age, 4)+rcs(educ, 6)+rcs(sal05, 4)+race, data = dat, subset = prog == 1)
  outcome0 = lm(sal07~rcs(age, 3)+rcs(educ, 8)+rcs(sal05, 5)+race, data = dat, subset = prog == 0)
  mu1 = mean(predict(outcome1, dat))
  mu0 = mean(predict(outcome0, dat))
  stand = mu1 - mu0
  return(stand)
}
B = 500
boot_stand = rep(NA, B)
for(b in 1:B){
  boot_sample = dat[sample(1:nrow(dat), replace = T), ]
  boot_stand[b] = get_stand(boot_sample)
}
saveRDS(boot_stand, "boot_stand.rds")
boot_stand = readRDS("boot_stand.rds")
```

```
stand.ci = round(quantile(boot_stand, c(0.025, 0.975)), 2)
par(mfrow = c(1, 2))
plot(residuals(outcome1)~fitted(outcome1), xlab = "fitted values", ylab = "residuals",
     main = "prog = 1")
abline(h = 0, col = 2)
plot(residuals(outcome0)~fitted(outcome0), xlab = "fitted values", ylab = "residuals",
     main = "prog = 0")
abline(h = 0, col = 2)
w = ifelse(dat$prog[dat$educ > 12] == 1, 1/propensities[dat$educ > 12],
           -1/(1-propensities[dat$educ > 12]))
iptw1 = mean(w*dat$sal07[dat$educ > 12])
w = ifelse(dat$prog[dat$educ <= 12] == 1, 1/propensities[dat$educ <= 12],
           -1/(1-propensities[dat$educ <= 12]))
iptw0 = mean(w*dat$sal07[dat$educ <= 12])
iptw_diff_educ = iptw1 - iptw0
get_iptw_diff_educ = function(dat){
  fit = glm(prog~rcs(age, 6)+educ+rcs(sal05, 6)+race, family = "binomial", data = dat)
  propensities = predict(fit, type = "response")
  w = ifelse(dat$prog[dat$educ > 12] == 1, 1/propensities[dat$educ > 12], -1/(1-propensities[dat$educ >
  iptw1 = mean(w*dat$sal07[dat$educ > 12])
  w = ifelse(dat$prog[dat$educ <= 12] == 1, 1/propensities[dat$educ <= 12], -1/(1-propensities[dat$educ
  iptw0 = mean(w*dat$sal07[dat$educ <= 12])
  return(iptw1 - iptw0)
}
B = 500
boot_iptw_diff_educ = rep(NA, B)
for(b in 1:B){
  boot_sample = dat[sample(1:nrow(dat), replace = T), ]
  boot_iptw_diff_educ[b] = get_iptw_diff_educ(boot_sample)
}
saveRDS(boot_iptw_diff_educ, "boot_iptw_diff_educ.rds")
boot_iptw_diff_educ = readRDS("boot_iptw_diff_educ.rds")
iptw_diff_educ.lower = round(quantile(boot_iptw_diff_educ, 0.05), 2)
get_iptw1 = function(dat){
  fit = glm(prog~rcs(age, 6)+educ+rcs(sal05, 6)+race, family = "binomial", data = dat)
  propensities = predict(fit, type = "response")
  w = ifelse(dat$prog[dat$educ > 12] == 1, 1/propensities[dat$educ > 12],
             -1/(1-propensities[dat$educ > 12]))
  iptw1 = mean(w*dat$sal07[dat$educ > 12])
  return(iptw1 - iptw0)
}

get_iptw0 = function(dat){
  fit = glm(prog~rcs(age, 6)+educ+rcs(sal05, 6)+race, family = "binomial", data = dat)
  propensities = predict(fit, type = "response")
  w = ifelse(dat$prog[dat$educ <= 12] == 1, 1/propensities[dat$educ <= 12],
             -1/(1-propensities[dat$educ <= 12]))
  iptw0 = mean(w*dat$sal07[dat$educ <= 12])
  return(iptw0)
}
B = 500
boot_iptw1 = boot_iptw0 = rep(NA, B)
for(b in 1:B){
```

```
  boot_sample = dat[sample(1:nrow(dat), replace = T), ]
  boot_iptw1[b] = get_iptw1(boot_sample)
  boot_iptw0[b] = get_iptw0(boot_sample)
}
saveRDS(boot_iptw1, "boot_iptw1.rds")
saveRDS(boot_iptw0, "boot_iptw0.rds")
boot_iptw1 = readRDS("boot_iptw1.rds")
iptw1.ci = round(quantile(boot_iptw1, c(0.025, 0.975)), 2)
boot_iptw0 = readRDS("boot_iptw0.rds")
iptw0.ci = round(quantile(boot_iptw0, c(0.025, 0.975)), 2)
```